

Main Project 02 – ICE HRM WEBSITE

Introduction

In this ICE HRM, which is used to get the human resource activities on real time actions. We are going to perform the load test, stress test, spike test, endurance test, and volume test here.

- Web application at <https://www.icehrm-open.gamonoid.com/>

Test Setup

ICE HRM

- | | | |
|--|---|--------------|
| 1. https://www.icehrm-open.gamonoid.com/ | - | ICE HRM |
| 2. https://www.icehrm-open.gamonoid.com/login.php | - | Login Page |
| 3. https://www.icehrm-open.gamonoid.com/service.php | - | Service Page |

Test-Setup

1. From blaze meter to get the data from website.
2. Using Synchronizing Timer to integrate no of user requests.
3. Using Random Controller to process the requests on randomly.
4. Using Listeners to get the data.
5. Using config elements to record used to provide default values when making a request to a server with samplers.

Environment Setup

1. **Processor** - 11th Gen Intel(R) Core(TM) i5-1135G7 @ 2.40GHz 2.42 GHz
2. **RAM** – 8GB

Jenkins

It integrated with Jenkin's.

GitHub

1. https://github.com/senthilraja29/Performance_Testing/blob/master/ice-HRM-Data-01%20-%20Load%20Test.jmx
2. https://github.com/senthilraja29/Performance_Testing/blob/master/ice-HRM-Data-01%20-%20Spike%20Test.jmx
3. https://github.com/senthilraja29/Performance_Testing/blob/master/ice-HRM-Data-01%20-%20Stress%20Test.jmx
4. https://github.com/senthilraja29/Performance_Testing/blob/master/ice-HRM-Data-01%20-%20Volume%20Test.jmx
5. https://github.com/senthilraja29/Performance_Testing/blob/master/ice-HRM-Data-01%20-%20Endurance%20Test.jmx

Environment

Running Environment JMeter.

Test Results

Load Test

In this load test to perform 500 concurrent users hitting at 31 mints by using concurrency thread group. After conducting load test around 69 thousand users hit the website within 31 mints, we received **0.16%** of error.

Summary Report

Name:Summary Report

Comments:

Write results to file / Read from file

Filenamenance testing\Projects\Main 02\JCE HRM\Load Test\jce-HRM-Data-01 - Load Test.csBrowse...Log/Display Only:☐ Errors☐ SuccessesConfigure

Label	# Samples	Average	Min	Max	Std. Dev.	Error %	Throughput	Received KB...	Sent KB/sec	Avg. Bytes
download	3883	4605	687	90854	4042.18	0.00%	2.1/sec	1.71	1.08	841.0
Attendance_...	3740	4679	664	90903	5381.55	0.05%	2.0/sec	1.63	1.14	829.3
travel	3846	9523	1916	94611	6405.11	0.13%	2.1/sec	16.37	2.31	8111.6
Personal_Inf...	3865	9729	1097	102152	7349.54	0.18%	2.1/sec	16.56	2.41	8162.0
Admin	3810	9763	1459	93798	7699.31	0.16%	2.0/sec	16.26	2.32	8127.5
admin_Man...	7537	9684	1350	117651	7393.78	0.17%	4.0/sec	32.09	4.55	8116.4
Finance	7823	9628	1337	93738	7239.95	0.22%	4.2/sec	33.32	4.72	8113.7
company_str...	3859	9519	1719	95005	6427.25	0.08%	2.1/sec	16.48	2.36	8138.6
Time_Mana...	7525	9713	1265	94831	7592.73	0.19%	4.0/sec	32.17	4.65	8145.4
Travel_Mana...	3925	9705	1475	93985	7377.83	0.18%	2.1/sec	16.79	2.42	8144.9
admin_Empl...	3916	9591	1477	94585	7147.99	0.20%	2.1/sec	16.71	2.38	8121.4
attendance	3757	9632	1370	94025	6908.32	0.24%	2.0/sec	16.01	2.27	8114.3
Basic_Inform...	3824	9823	1142	94711	8032.00	0.24%	2.1/sec	16.36	2.38	8151.7
User_Reports	3828	9633	1293	93809	7060.16	0.03%	2.1/sec	16.36	2.34	8141.4
login	3894	9858	1923	93984	6825.51	0.18%	2.1/sec	16.57	2.62	8104.0
TOTAL	69032	9120	664	117651	7185.17	0.16%	37.1/sec	265.20	39.94	7323.8

Aggregate Report

Name:

Aggregate Report

Comments:

Write results to file / Read from file

Filename

Browse...

Log/Display Only:

☐ Errors

☐ Successes

Configure

Label	# Samples	Average	Median	90% Line	95% Line	99% Line	Min	Maximum	Error %	Throughp...	Received ...	Sent KB/sec
download	3883	4605	4348	7361	8276	9741	687	90854	0.00%	2.1/sec	1.71	1.08
Attendanc...	3740	4679	4185	7472	8331	9985	664	90903	0.05%	2.0/sec	1.63	1.14
travel	3846	9523	9085	11528	11984	14519	1916	94611	0.13%	2.1/sec	16.37	2.31
Personal_I...	3865	9729	9191	11554	11988	16607	1097	102152	0.18%	2.1/sec	16.56	2.41
Admin	3810	9763	9107	11581	12029	15842	1459	93798	0.16%	2.0/sec	16.26	2.32
admin_M...	7537	9684	9098	11575	12004	16579	1350	117651	0.17%	4.0/sec	32.09	4.55
Finance	7823	9628	9030	11511	11972	14909	1337	93738	0.22%	4.2/sec	33.32	4.72
company_...	3859	9519	9067	11528	11957	14083	1719	95005	0.08%	2.1/sec	16.48	2.36
Time_Ma...	7525	9713	9101	11558	11992	17170	1265	94831	0.19%	4.0/sec	32.17	4.65
Travel_Ma...	3925	9705	9140	11583	12001	15222	1475	93985	0.18%	2.1/sec	16.79	2.42
admin_E...	3916	9591	9023	11508	11978	14328	1477	94585	0.20%	2.1/sec	16.71	2.38
attendance	3757	9632	9144	11582	11993	15464	1370	94025	0.24%	2.0/sec	16.01	2.27
User_Rep...	3828	9633	9095	11519	11961	14971	1293	93809	0.03%	2.1/sec	16.36	2.34
Basic_Info...	3824	9823	9049	11580	12073	23831	1142	94711	0.24%	2.1/sec	16.36	2.38
login	3894	9858	9358	11634	12048	14100	1923	93984	0.18%	2.1/sec	16.57	2.62
TOTAL	69032	9120	8778	11467	11939	14865	664	117651	0.16%	37.1/sec	265.20	39.94



Row Labels	StdDev of Latency	Max of Latency	Average of Latency
Admin	5854.925826	91073	4788.622
admin_Employees	5590.41882	90968	4680.299
admin_Employees-0	5592.037243	90968	4682.419
admin_Employees-1	4981.605808	89645	4796.213
admin_Manage	5646.105638	90900	4762
admin_Manage-0	5647.909608	90900	4764.861
admin_Manage-1	5257.25668	89944	4804.509
Admin-0	5854.925826	91073	4788.622
Admin-1	5592.381753	89907	4850.149
attendance	5385.923167	93165	4748.714
Attendance_Reportdownload	5379.893273	90903	4657.462
attendance-0	5387.121722	93165	4751.485
attendance-1	4883.324862	89166	4767.86
Basic_Information	5852.239846	90831	4763.169
Basic_Information-0	5853.996507	90831	4766.99
Basic_Information-1	6023.528231	89586	4933.592
company_structure	4753.293401	90908	4617.783
company_structure-0	4753.713086	90908	4618.479
company_structure-1	4827.233796	89627	4787.815
download	4043.422983	90854	4582.075
Finance	5354.488222	90976	4672.562
Finance-0	5356.082186	90976	4675.271
Finance-1	5344.520371	89872	4829.022
login	4973.639725	90914	5100.401
login-0	4974.398904	90914	5102.029
login-1	5159.932531	89413	4647.311
Personal_Information	5668.507486	90859	4801.848
Personal_Information-0	5668.714544	90859	4803.091
Personal_Information-1	5026.137786	89911	4779.936
Time_Management	5719.542349	90957	4744.838
Time_Management-0	5720.964388	90957	4747.524
Time_Management-1	5528.093691	89976	4843.239
travel	4529.77454	90838	4611.088
Travel_Management	5763.775717	90953	4759.213
Travel_Management-0	5764.799658	90953	4762.363
Travel_Management-1	5157.578677	89725	4825.751
travel-0	4530.156952	90838	4611.786
travel-1	5022.038738	89959	4803.999
User_Reports	4709.724324	90892	4647.643
User_Reports-0	4709.724324	90892	4647.643
User_Reports-1	5721.843155	90088	4872.939
Grand Total	5358.601043	93165	4760.87

Observations:

1. **0.16%** errors encountered during load test around 69k transactions.
2. Average Response time **9120 ms**
3. 95th percentile fall under – **11939 ms**
4. Throughput (requests per second) - **37.1/ sec**
5. Network performance (Max latency) – **93165 ms**
6. Network performance (Average latency) – **4760.87 ms**

Stress Test

In this stress test, which is conducted by stepping thread groups to perform 1500 users holds for 700 seconds. In this load, 300 threads are gradually added by each 20-second interval.

Summary Report

Name:

Summary Report

Comments:

Write results to file / Read from file

Filename

nce testing\Projects\Main 02\ICE HRM\Stress Test\ice-HRM-Data-01 - Stress Test.csv

Browse...

Log/Display Only:

☐ Errors

☐ Successes

Configure

Label	# Samples	Average	Min	Max	Std. Dev.	Error %	Throughput	Received KB...	Sent KB/sec	Avg. Bytes
download	2921	5019	479	47847	7214.22	57.45%	3.8/sec	3.31	1.89	894.8
Attendance...	2818	5115	504	47863	7552.40	57.38%	3.6/sec	3.18	1.99	893.2
Basic_Inform...	2772	7179	623	43174	7774.50	61.08%	3.6/sec	13.23	2.85	3773.9
company_str...	2791	7228	605	47845	8370.28	63.31%	3.6/sec	12.76	2.77	3615.0
Finance	5625	7393	630	47860	8309.43	61.88%	7.3/sec	26.46	5.59	3719.2
Admin	2848	7357	635	47833	8254.30	61.76%	3.7/sec	13.42	2.85	3727.5
Time_Mana...	5630	7259	629	47823	8038.05	61.62%	7.3/sec	26.62	5.74	3739.2
admin_Empl...	2873	7421	637	47849	8185.25	60.84%	3.7/sec	13.78	2.88	3793.1
Personal_Inf...	2821	7646	629	47828	8536.69	60.62%	3.7/sec	13.69	2.92	3837.7
User_Reports	2908	7450	620	47853	8258.69	61.24%	3.8/sec	13.87	2.94	3771.5
attendance	2770	7274	612	43179	7996.52	61.52%	3.6/sec	13.09	2.77	3736.7
travel	2824	7224	639	43337	8453.75	63.70%	3.7/sec	12.79	2.74	3580.6
login	2809	7146	656	43177	7789.48	62.41%	3.6/sec	12.88	3.23	3625.1
admin_Man...	5739	7231	611	47860	8173.74	62.94%	7.4/sec	26.43	5.67	3642.4
Travel_Mana...	2873	7307	615	47863	8089.49	61.33%	3.7/sec	13.66	2.94	3761.1
TOTAL	51022	7060	479	47863	8121.87	61.41%	66.1/sec	219.17	49.80	3396.9

Aggregate Report

Name:

Aggregate Report

Comments:

Write results to file / Read from file

Filename

Browse...

Log/Display Only:

☐ Errors
 ☐ Successes

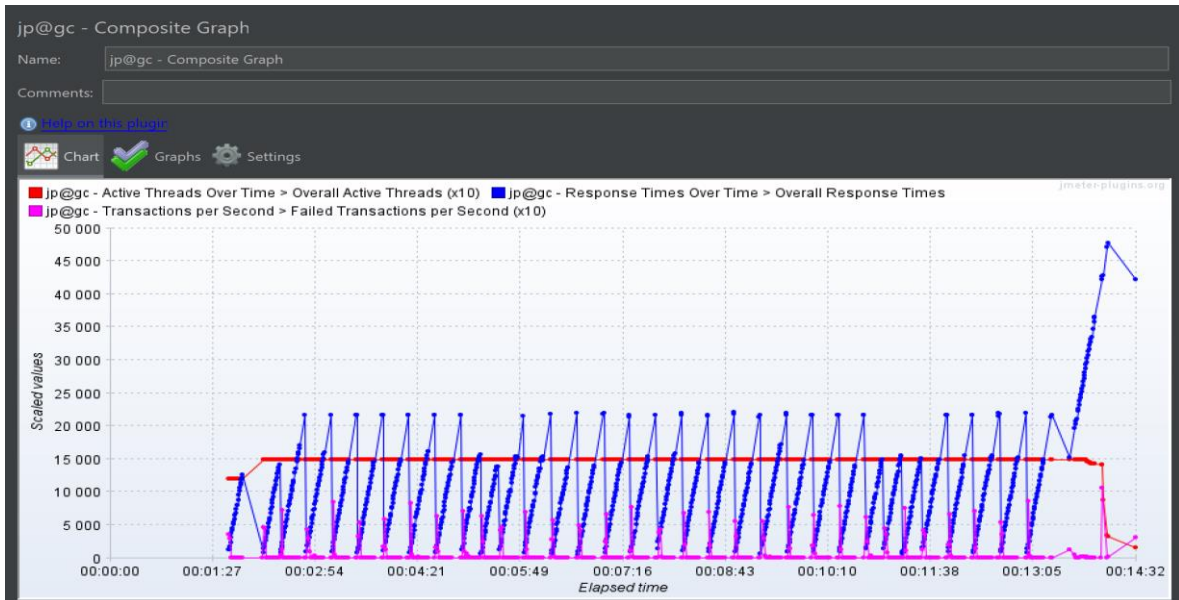
Configure

Label	# Samples	Average	Median	90% Line	95% Line	99% Line	Min	Maximum	Error %	Throughp...	Received ...	Sent KB/sec
download	2921	5019	2413	9576	12649	42902	479	47847	57.45%	3.8/sec	3.31	1.89
Attendanc...	2818	5115	2420	9511	12867	42940	504	47863	57.38%	3.6/sec	3.18	1.99
Basic_Info...	2772	7179	2619	14557	15544	42882	623	43174	61.08%	3.6/sec	13.23	2.85
company_...	2791	7228	2475	14588	15890	42908	605	47845	63.31%	3.6/sec	12.76	2.77
Finance	5625	7393	2599	14554	15691	42907	630	47860	61.88%	7.3/sec	26.46	5.59
Admin	2848	7357	2603	14521	15522	42908	635	47833	61.76%	3.7/sec	13.42	2.85
Time_Ma...	5630	7259	2534	14599	15623	42904	629	47823	61.62%	7.3/sec	26.62	5.74
admin_E...	2873	7421	2603	14572	15799	42897	637	47849	60.84%	3.7/sec	13.78	2.88
Personal_J...	2821	7646	2623	14612	16102	42935	629	47828	60.62%	3.7/sec	13.69	2.92
User_Rep...	2908	7450	2623	14524	15651	42904	620	47853	61.24%	3.8/sec	13.87	2.94
attendance	2770	7274	2592	14502	15507	42907	612	43179	61.52%	3.6/sec	13.09	2.77
travel	2824	7224	2449	14448	16025	42928	639	43337	63.70%	3.7/sec	12.79	2.74
login	2809	7146	2520	14521	15617	42874	656	43177	62.41%	3.6/sec	12.88	3.23
admin_M...	5739	7231	2490	14531	15578	42914	611	47860	62.94%	7.4/sec	26.43	5.67
Travel_Ma...	2873	7307	2578	14625	15647	42948	615	47863	61.33%	3.7/sec	13.66	2.94
TOTAL	51022	7060	2525	14455	15594	42907	479	47863	61.41%	66.1/sec	219.17	49.80

☐ Include group name in label

Save Table Data

☒ Save Table Header



Row Labels	StdDev of Latency	Max of Latency	Average of Latency
Admin	3151.801385	21421	3701.491
admin_Employees	3249.589438	21516	3731.566
admin_Employees-0	3506.078621	21516	6182.593
admin_Employees-1	2352.446352	10091	5199.785
admin_Manage	3218.842805	21951	3700.017
admin_Manage-0	3476.188132	21951	6233.898
admin_Manage-1	2406.333038	10093	5028.898
Admin-0	3357.295583	21421	6164.959
Admin-1	2324.791504	10064	5138.927
attendance	3224.355125	21799	3775.666
Attendance_Reportdownload	3099.257993	21567	3623.988
attendance-0	3418.874026	21799	6268.75
attendance-1	2366.325078	10206	5074.635
Basic_Information	3214.901873	21805	3763.669
Basic_Information-0	3439.464513	21805	6236.339
Basic_Information-1	2323.198995	10126	5165.366
company_structure	3182.46453	21752	3660.197
company_structure-0	3468.76868	21752	6210.368
company_structure-1	2349.510581	10064	5069.574
download	3099.034722	21411	3651.229
Finance	3197.238048	22003	3706.646
Finance-0	3441.528712	22003	6147.943
Finance-1	2380.37704	10146	5098.328
login	3232.242849	21557	3837.402
login-0	3268.752984	21557	6593.177
login-1	2207.281034	10142	5128.546
Personal_Information	3388.052262	21494	3832.798
Personal_Information-0	3623.654925	21494	6341.098
Personal_Information-1	2366.951402	10055	4989.765
Time_Management	3262.56757	21786	3755.379
Time_Management-0	3503.117191	21786	6270.415
Time_Management-1	2362.287012	10038	5112.053
travel	3200.097058	21860	3576.114
Travel_Management	3330.288978	21503	3817.618
Travel_Management-0	3567.246542	21503	6341.678
Travel_Management-1	2364.339741	10079	5041.348
travel-0	3602.516829	21860	6075.498
travel-1	2383.446194	9607	5119.945
User_Reports	3264.149807	22102	3764.918
User_Reports-0	3498.589793	22102	6231.872
User_Reports-1	2373.530419	10046	5104.491
Grand Total	3283.433382	22102	4571.722

Observations:

1. During stress test error occurred which is **61.41%**.
2. Average Response time **7060ms**
3. 95th percentile falls under - **15594ms**
4. Throughput (requests per second) – **66.1/Sec**
5. Network performance (Max latency) – **22102ms**
6. Network performance (Average latency) – **4571.72ms**

Spike Test

In this spike test, which is conducted by open model thread group to perform 800 users interact within 40 seconds. In this test started by 10 seconds and completed by 60 seconds and maintained 1 minute of transaction rate up to 100. Load rate will be 300/seconds. Around 7080 samples were transitioned.

Summary Report

Name:

Summary Report

Comments:

Write results to file / Read from file

Filename

ance testing\Projects\Main 02\ICE HRM\Spike Test\ice-HRM-Data-01 - Spike Test.cs

Browse...

Log/Display Only:

☐ Errors

☐ Successes

Configure

Label	# Samples	Average	Min	Max	Std. Dev.	Error %	Throughput	Received KB...	Sent KB/sec	Avg. Bytes
download	377	2445	5	24139	3008.46	63.40%	3.2/sec	2.53	1.63	820.2
Attendance_...	375	2091	7	26145	2772.51	68.00%	3.1/sec	2.49	1.77	812.9
attendance	401	3432	345	32382	4975.40	73.07%	3.5/sec	9.71	2.52	2803.6
admin_Empl...	384	4313	336	29038	5365.49	64.58%	3.4/sec	11.34	2.58	3418.6
User_Reports	380	4313	339	28829	5294.00	68.16%	3.8/sec	11.85	2.90	3194.3
Travel_Mana...	377	4175	347	24850	5169.49	66.84%	4.2/sec	13.56	3.27	3285.4
Finance	734	3916	340	31785	5231.37	70.44%	6.5/sec	18.94	4.70	2997.1
company_str...	382	3354	336	25195	4752.62	73.56%	3.4/sec	9.16	2.43	2777.7
admin_Man...	818	4042	7	21677	5034.16	68.09%	6.8/sec	21.27	5.12	3184.8
Time_Mana...	852	3822	335	32406	5078.38	70.54%	8.5/sec	25.04	6.35	3002.0
Basic_Inform...	387	3886	340	21683	4914.33	68.22%	3.4/sec	10.60	2.60	3172.1
Admin	423	3955	7	15933	4935.87	69.03%	3.5/sec	10.78	2.64	3121.6
travel	381	3319	329	15353	4689.06	75.33%	3.4/sec	8.69	2.34	2638.8
Personal_Inf...	411	4353	7	15372	5154.97	64.23%	3.4/sec	11.64	2.69	3469.1
login	398	4138	5	22053	5299.17	70.35%	3.3/sec	9.69	2.86	2981.3
TOTAL	7080	3749	5	32406	4929.34	69.05%	59.2/sec	164.09	43.11	2838.0

Aggregate Report

Name: Aggregate Report

Comments:

Write results to file / Read from file

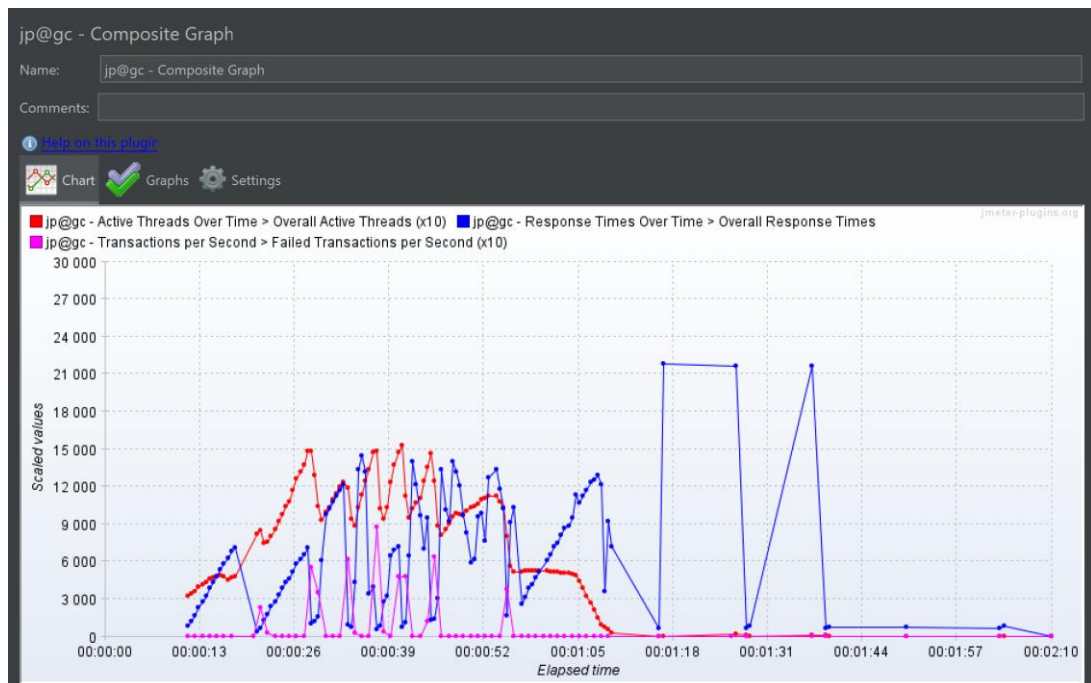
Filename

Browse...

Log/Display Only: ☐ Errors ☐ Successes

Configure

Label	# Samples	Average	Median	90% Line	95% Line	99% Line	Min	Maximum	Error %	Throughp...	Received ...	Sent KB/sec
download	377	2445	703	7092	7983	9334	5	24139	63.40%	3.2/sec	2.53	1.63
Attendanc...	375	2091	584	6514	7537	9246	7	26145	68.00%	3.1/sec	2.49	1.77
attendance	401	3432	587	11928	13184	14732	345	32382	73.07%	3.5/sec	9.71	2.52
admin_E...	384	4313	743	12408	13264	14765	336	29038	64.58%	3.4/sec	11.34	2.58
User_Rep...	380	4313	764	12550	13450	14928	339	28829	68.16%	3.8/sec	11.85	2.90
Travel_Ma...	377	4175	689	12443	13160	14918	347	24850	66.84%	4.2/sec	13.56	3.27
Finance	734	3916	693	12326	13494	14940	340	31785	70.44%	6.5/sec	18.94	4.70
company_...	382	3354	602	12093	12940	14665	336	25195	73.56%	3.4/sec	9.16	2.43
admin_M...	818	4042	701	12219	13566	14797	7	21677	68.09%	6.8/sec	21.27	5.12
Time_Ma...	852	3822	705	12238	13206	15037	335	32406	70.54%	8.5/sec	25.04	6.35
Basic_Inf...	387	3886	737	12070	13046	14703	340	21683	68.22%	3.4/sec	10.60	2.60
Admin	423	3955	718	12254	13142	15001	7	15933	69.03%	3.5/sec	10.78	2.64
travel	381	3319	559	11861	13250	14170	329	15353	75.33%	3.4/sec	8.69	2.34
Personal_I...	411	4353	722	12136	13280	14700	7	15372	64.23%	3.4/sec	11.64	2.69
login	398	4138	709	12849	14086	15385	5	22053	70.35%	3.3/sec	9.69	2.86
TOTAL	7080	3749	700	12080	13216	14907	5	32406	69.05%	59.2/sec	164.09	43.11



Row Labels	StdDev of Latency	Max of Latency	Average of Latency
Admin	2784.647154	10209	2384.856
admin_Employees	3214.575468	26310	2500.159
admin_Employees-0	3391.548814	26310	5603.082
admin_Employees-1	2361.621945	8428	4768.267
admin_Manage	2872.346339	21420	2389.477
admin_Manage-0	2618.176966	21420	5482.921
admin_Manage-1	2638.772836	8544	4444.362
Admin-0	2353.083872	10209	5555.85
Admin-1	2565.179204	8431	4334.007
attendance	2931.4915	28672	1999.803
Attendance_Reportdownload	2776.123182	26145	2091.115
attendance-0	3555.775695	28672	5282.574
attendance-1	2594.840555	8498	4701.762
Basic_Information	2829.590007	21431	2300.941
Basic_Information-0	2733.809823	21431	5413.796
Basic_Information-1	2475.719668	8536	4460.788
company_structure	2855.213547	24644	2082.165
company_structure-0	3097.720566	24644	5504.094
company_structure-1	2575.746088	8314	4148.137
download	3012.415784	24139	2444.968
Finance	3122.319526	24762	2347.23
Finance-0	3184.076003	24762	5913.225
Finance-1	2557.479024	8578	4715.582
login	3153.50357	21797	2601.485
login-0	2596.69876	21797	6400.277
login-1	2657.082553	7915	4463.511
Personal_Information	2827.266454	10131	2500.428
Personal_Information-0	2284.38325	10131	5548.069
Personal_Information-1	2512.41675	8519	4756.35
Time_Management	3008.180596	28689	2304.586
Time_Management-0	3042.55612	28689	5733.182
Time_Management-1	2572.705965	8536	4530.453
travel	2665.7514	10119	2058.74
Travel_Management	3099.800651	24287	2471.355
Travel_Management-0	3066.487591	24287	5559.8
Travel_Management-1	2561.380313	8538	4422.441
travel-0	2232.542302	10119	5807.716
travel-1	2626.399188	8238	4403.734
User_Reports	3131.730654	26211	2616.945
User_Reports-0	2863.064119	26211	5894.09
User_Reports-1	2672.765484	8693	4441.497
Grand Total	3193.329441	28689	3398.096

Observations:

1. During Spike test error occurred which is **69.05%**
2. Average Response time **3749ms**
3. 95th percentile Response falls under – **132216ms**
4. Throughput (requests per second) – **59.2/sec**
5. Network performance (Max latency) – **28689ms**
6. Network performance (Average latency) – **3398.09ms**

Endurance Test

In this endurance test, which is conducted by Free from Arrivals thread group to perform 300 users interact within 61 minutes. In this test, the full load started after 65 seconds and was completed by 1 hour 01 minutes. All the requests will under synchronize timer for 300 concurrent users.

Summary Report

Name:

Summary Report

Comments:

Write results to file / Read from file

Filename

ng\Projects\Main 02\JCE HRM\Endurance Test\jce-HRM-Data-01 - Endurance Test.c

Browse...

Log/Display Only:

☐ Errors

☐ Successes

Configure

Label	# Samples	Average	Min	Max	Std. Dev.	Error %	Throughput	Received KB...	Sent KB/sec	Avg. Bytes
Attendance_...	5463	2859	417	16242	1471.30	0.00%	1.5/sec	1.21	0.85	829.7
download	5376	3019	406	24843	1551.42	0.02%	1.5/sec	1.21	0.76	841.2
attendance	5393	5965	1142	23504	1128.36	0.00%	1.5/sec	11.69	1.66	8127.9
Admin	5356	5981	1066	10702	1077.64	0.00%	1.5/sec	11.62	1.66	8134.6
admin_Empl...	5508	5974	1156	22378	1159.23	0.00%	1.5/sec	11.95	1.70	8134.6
Personal_Inf...	5570	5987	1165	24877	1145.80	0.00%	1.5/sec	12.14	1.77	8170.7
company_str...	5443	5956	882	24969	1172.53	0.00%	1.5/sec	11.83	1.69	8143.0
admin_Man...	10697	6034	824	732582	7120.89	0.01%	2.9/sec	23.19	3.29	8126.1
Time_Mana...	10689	5970	736	28964	1208.43	0.00%	2.9/sec	23.26	3.36	8156.9
travel	5451	5982	840	24586	1226.64	0.00%	1.5/sec	11.81	1.66	8118.1
Travel_Mana...	5430	5967	1131	22425	1163.71	0.00%	1.5/sec	11.81	1.71	8155.2
Basic_Inform...	5438	5975	1035	28333	1301.18	0.02%	1.5/sec	11.85	1.72	8163.8
Finance	10820	5967	1105	27976	1194.56	0.01%	3.0/sec	23.46	3.33	8125.8
User_Reports	5451	5959	1095	24626	1128.16	0.00%	1.5/sec	11.84	1.70	8142.9
login	5457	6225	1216	23459	1046.30	0.00%	1.5/sec	11.81	1.87	8113.6
TOTAL	97542	5655	406	732582	2794.27	0.00%	26.6/sec	190.69	28.72	7327.3

Aggregate Report

Name:

Aggregate Report

Comments:

Write results to file / Read from file

Filename

Browse...

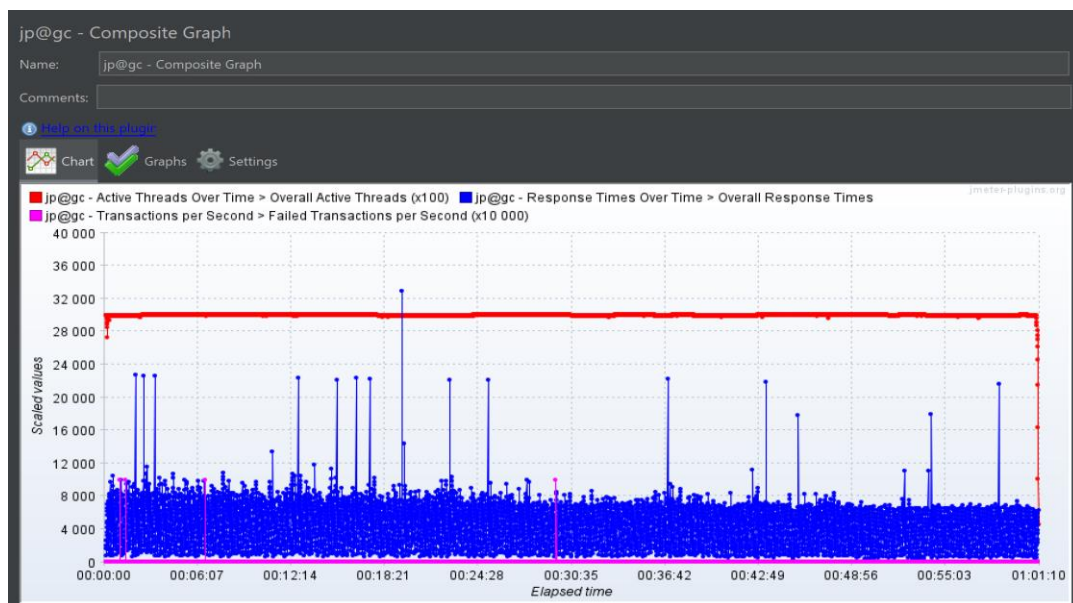
Log/Display Only:

☐ Errors

☐ Successes

Configure

Label	# Samples	Average	Median	90% Line	95% Line	99% Line	Min	Maximum	Error %	Throughp...	Received ...	Sent KB/sec
Attendanc...	5463	2859	2643	4969	5408	6341	417	16242	0.00%	1.5/sec	1.21	0.85
download	5376	3019	2924	5015	5498	6529	406	24843	0.02%	1.5/sec	1.21	0.76
attendance	5393	5965	6089	7100	7390	8478	1142	23504	0.00%	1.5/sec	11.69	1.66
Admin	5356	5981	6107	7123	7431	8484	1066	10702	0.00%	1.5/sec	11.62	1.66
admin_E...	5508	5974	6086	7127	7435	8614	1156	22378	0.00%	1.5/sec	11.95	1.70
Personal_I...	5570	5987	6092	7111	7428	8455	1165	24877	0.00%	1.5/sec	12.14	1.77
company_...	5443	5956	6059	7110	7400	8253	882	24969	0.00%	1.5/sec	11.83	1.69
admin_M...	10697	6034	6078	7097	7402	8475	824	732582	0.01%	2.9/sec	23.19	3.29
Time_Ma...	10689	5970	6073	7128	7426	8480	736	28964	0.00%	2.9/sec	23.26	3.36
travel	5451	5982	6085	7109	7404	8485	840	24586	0.00%	1.5/sec	11.81	1.66
Travel_Ma...	5430	5967	6075	7127	7421	8545	1131	22425	0.00%	1.5/sec	11.81	1.71
Basic_Info...	5438	5975	6084	7116	7419	8517	1035	28333	0.02%	1.5/sec	11.85	1.72
Finance	10820	5967	6086	7118	7403	8478	1105	27976	0.01%	3.0/sec	23.46	3.33
User_Rep...	5451	5959	6092	7121	7394	8426	1095	24626	0.00%	1.5/sec	11.84	1.70
login	5457	6225	6298	7179	7463	8744	1216	23459	0.00%	1.5/sec	11.81	1.87
TOTAL	97542	5655	5947	7073	7365	8430	406	732582	0.00%	26.6/sec	190.69	28.72



Row Labels	StdDev of Latency	Max of Latency	Average of Latency
Admin	1462.714942	9590	3022.622
admin_Employees	1535.577241	21862	3034.778
admin_Employees-0	1535.577241	21862	3034.778
admin_Employees-1	760.6367733	7732	2905.934
admin_Manage	1513.476506	22100	3015.319
admin_Manage-0	1513.266367	22100	3015.601
admin_Manage-1	7074.320719	730476	2977.905
Admin-0	1462.714942	9590	3022.622
Admin-1	761.2718591	6764	2920.58
attendance	1504.974421	21890	3006.391
Attendance_Reportdownload	1467.378201	16242	2850.345
attendance-0	1504.974421	21890	3006.391
attendance-1	745.8716438	6093	2924.841
Basic_Information	1635.265461	25845	3031.958
Basic_Information-0	1634.531599	25845	3033.073
Basic_Information-1	758.0015566	6091	2904.316
company_structure	1569.509648	21848	3010.571
company_structure-0	1569.509648	21848	3010.571
company_structure-1	770.8410446	6056	2914.07
download	1520.489489	21471	3006.411
Finance	1543.928311	25155	3028.924
Finance-0	1543.724986	25155	3029.204
Finance-1	768.8525948	6410	2901.332
login	1439.421019	21918	3373.349
login-0	1439.421019	21918	3373.349
login-1	721.6353244	6159	2814.977
Personal_Information	1540.579355	21978	3038.513
Personal_Information-0	1540.579355	21978	3038.513
Personal_Information-1	760.2350455	6064	2912.329
Time_Management	1582.360815	27479	3029.543
Time_Management-0	1582.360815	27479	3029.543
Time_Management-1	755.461449	6278	2903.061
travel	1612.003842	21926	3047.069
Travel_Management	1555.746018	21907	3026.597
Travel_Management-0	1555.746018	21907	3026.597
Travel_Management-1	757.9274323	6097	2905.55
travel-0	1611.242232	21926	3048.187
travel-1	764.5832006	6101	2897.45
User_Reports	1501.051789	21565	3038.392
User_Reports-0	1501.051789	21565	3038.392
User_Reports-1	746.0324935	6102	2891.749
Grand Total	1937.43899	730476	2999.502

Observations:

1. During Endurance test results shows Null error around 97k transactions.
2. Average Response time **5655ms**
3. 95th percentile Response times - **7365ms**
4. Throughput (requests per second) – **26.6/Sec**
5. Network performance (Max latency) – **730476ms**
6. Network performance (Average latency) – **2999.50ms**

Volume Test

In this volume test, which is conducted by the Ultimate thread group to perform 1000 users interact with 60 mints. In this test, which was performed by various load actives 100, 500 and full load 1000. This full load 1000 users in 12 minutes. Each request is integrated with a random controller and random controller to simulate real-world scenarios.

Summary Report

Name:

Summary Report

Comments:

Write results to file / Read from file

Filename

testing\Projects\Main 02\ICE HRM\Volume Test\ice-HRM-Data-01 - Volume Test.c

Browse...

Log/Display Only:

☐ Errors

☐ Successes

Configure

Label	# Samples	Average	Min	Max	Std. Dev.	Error %	Throughput	Received KB...	Sent KB/sec	Avg. Bytes
Attendance_...	4497	3578	330	76202	4987.33	30.75%	2.5/sec	1.97	1.39	820.3
download	4525	3716	329	76216	5110.80	31.36%	2.5/sec	2.01	1.29	827.9
travel	4482	6976	330	79553	6900.68	32.75%	2.5/sec	13.79	2.30	5739.1
User_Reports	4475	6874	324	79391	6759.19	33.25%	2.5/sec	13.72	2.33	5715.3
Finance	8936	6990	328	79490	7113.33	33.35%	4.9/sec	27.31	4.60	5699.7
attendance	4532	6878	329	79404	7055.39	33.85%	2.5/sec	13.77	2.33	5665.1
company_str...	4569	6749	328	79503	6441.16	34.12%	2.5/sec	13.85	2.37	5653.0
admin_Empl...	4557	6941	329	79275	7225.69	34.43%	2.5/sec	13.74	2.34	5629.3
admin_Man...	9007	6865	327	103411	6862.44	33.36%	4.9/sec	27.36	4.61	5699.7
Admin	4488	7002	328	79668	7570.69	33.82%	2.5/sec	13.64	2.31	5668.6
Time_Mana...	8926	6975	322	79400	6982.21	33.00%	4.9/sec	27.40	4.69	5745.2
Personal_Inf...	4492	6551	327	78790	6075.46	35.69%	2.5/sec	13.38	2.35	5554.9
Basic_Inform...	4431	6863	330	79134	6712.61	33.13%	2.4/sec	13.62	2.34	5741.5
Travel_Mana...	4333	6935	327	79422	6986.61	33.03%	2.4/sec	13.34	2.28	5742.5
login	4440	7006	355	103328	7685.08	35.11%	2.4/sec	13.14	2.56	5543.4
TOTAL	80690	6538	322	103411	6857.77	33.37%	44.0/sec	221.04	39.91	5140.3

Aggregate Report

Name:

Aggregate Report

Comments:

Write results to file / Read from file

Filename

Browse...

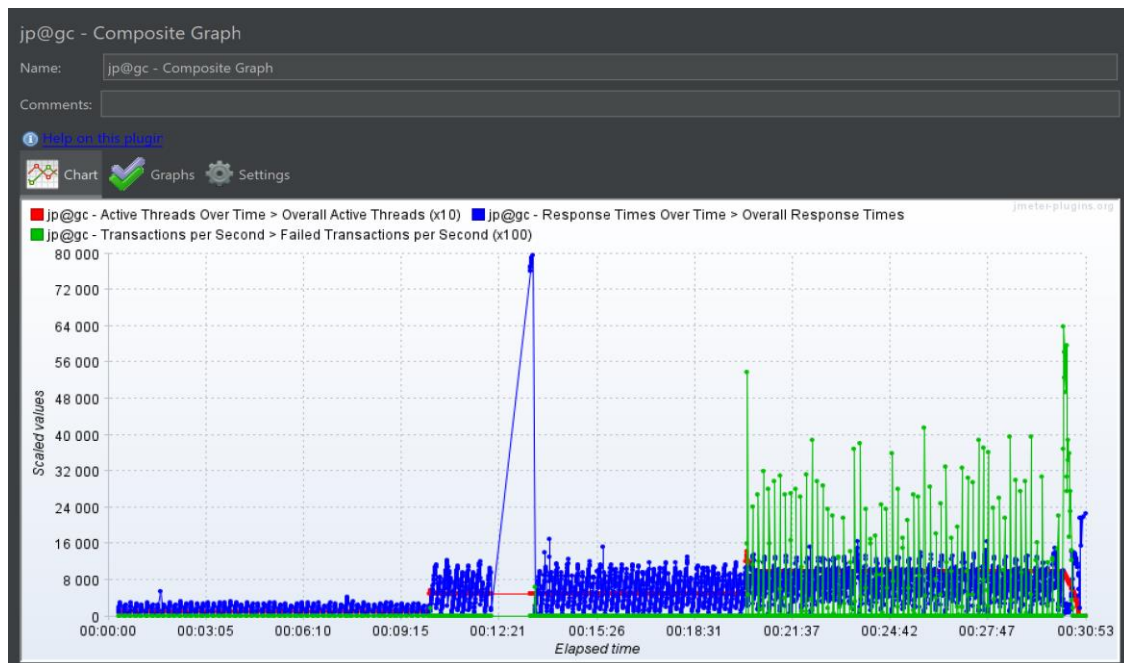
Log/Display Only:

☐ Errors

☐ Successes

Configure

Label	# Samples	Average	Median	90% Line	95% Line	99% Line	Min	Maximum	Error %	Throughp...	Received ...	Sent KB/sec
Attendanc...	4497	3578	2544	7215	8203	9763	330	76202	30.75%	2.5/sec	1.97	1.39
download	4525	3716	2679	7387	8219	9723	329	76216	31.36%	2.5/sec	2.01	1.29
travel	4482	6976	8303	11989	12884	16654	330	79553	32.75%	2.5/sec	13.79	2.30
User_Rep...	4475	6874	8268	11990	12863	15719	324	79391	33.25%	2.5/sec	13.72	2.33
Finance	8936	6990	8327	12017	12925	16451	328	79490	33.35%	4.9/sec	27.31	4.60
attendance	4532	6878	8168	12004	12961	15565	329	79404	33.85%	2.5/sec	13.77	2.33
company_...	4569	6749	8185	11883	12814	14879	328	79503	34.12%	2.5/sec	13.85	2.37
admin_E...	4557	6941	8221	12164	13015	16986	329	79275	34.43%	2.5/sec	13.74	2.34
admin_M...	9007	6865	8176	11993	12898	15543	327	103411	33.36%	4.9/sec	27.36	4.61
Admin	4488	7002	8254	11994	12898	23869	328	79668	33.82%	2.5/sec	13.64	2.31
Time_Ma...	8926	6975	8291	12005	12998	16567	322	79400	33.00%	4.9/sec	27.40	4.69
Personal_I...	4492	6551	7991	11876	12847	14508	327	78790	35.69%	2.5/sec	13.38	2.35
Basic_Info...	4431	6863	8269	12002	12878	15657	330	79134	33.13%	2.4/sec	13.62	2.34
Travel_Ma...	4333	6935	8312	11982	13004	15860	327	79422	33.03%	2.4/sec	13.34	2.28
login	4440	7006	8294	12363	13115	16631	355	103328	35.11%	2.4/sec	13.14	2.56
TOTAL	80690	6538	7399	11848	12816	15657	322	103411	33.37%	44.0/sec	221.04	39.91



Row Labels	StdDev of Latency	Max of Latency	Average of Latency
Admin	6249.528573	76186	3954.51
admin_Employees	6001.356421	76223	3925.956
admin_Employees-0	6846.992597	76223	5290.681
admin_Employees-1	2951.528517	74612	4436.814
admin_Manage	4990.947504	76229	3703.292
admin_Manage-0	5573.005192	76229	4925.966
admin_Manage-1	4084.111845	75274	4605.691
Admin-0	7162.11558	76186	5342.692
Admin-1	3445.548323	74851	4489.953
attendance	5671.230577	76180	3802.19
Attendance_Reportdownload	4987.957512	76202	3578.467
attendance-0	6433.290782	76180	5093.999
attendance-1	3199.132865	75272	4508.994
Basic_Information	5443.719158	76229	3832.575
Basic_Information-0	6124.46525	76229	5127.398
Basic_Information-1	2656.547505	74250	4408.784
company_structure	4881.824333	76156	3693.902
company_structure-0	5471.013513	76156	4985.188
company_structure-1	3181.311339	74857	4513.633
download	5111.412329	76216	3715.786
Finance	5661.773143	76183	3857.081
Finance-0	6406.107955	76183	5160.128
Finance-1	3436.194123	75135	4570.044
login	6338.998861	76204	4129.522
login-0	7260.032363	76204	5695.619
login-1	3259.54536	74985	4322.918
Personal_Information	4327.12327	76167	3582.585
Personal_Information-0	4791.794575	76167	4902.431
Personal_Information-1	3218.19516	75073	4499.108
Time_Management	5606.979215	76220	3859.677
Time_Management-0	6311.48884	76220	5136.436
Time_Management-1	3191.857183	75175	4524.031
travel	5317.451881	76183	3845.653
Travel_Management	5468.804832	76203	3828.535
Travel_Management-0	6148.928781	76203	5105.221
Travel_Management-1	3486.559992	75191	4510.588
travel-0	5948.268198	76183	5108.853
travel-1	3651.244947	75014	4517.599
User_Reports	5491.49091	76175	3807.939
User_Reports-0	6195.281088	76175	5106.825
User_Reports-1	2624.996961	74921	4482.738
Grand Total	5235.897107	76229	4362.792

Observations:

1. During Volume test error occurred which is **33.37%**.
2. Average Response time **6538ms**
3. 95th percentile Response times – **12816ms**
4. Throughput (requests per second) – **44.0/Sec**
5. Network performance (Max latency) – **76229ms**
6. Network performance (Average latency) – **4362.79ms**

Conclusion

After conducting various tests from various thread groups, we will come to know

- **Load Test** - After performing load test we received **0.16%** of error around 69k transactions. Error starts after 9 mins requests processed.
- **Stress Test** - After performing stress test we received **61.41%** of error around 51k transactions. Error starts at starting stage itself and requests processed Latency Report shows 21k maximum duration takes.
- **Endurance Test** - After performing Endurance test, we received **0.00%** of error around 97k transactions.
- **Spike Test** - After performing Spike test, we received **69.05%** of error around 7k transactions.
- **Volume Test** - After performing the volume test, an error occurred, which is **33.37%** of error around 81k transactions

Website can able to handle Endurance and Load Tests. This Webpage unable to handle stress test, volume and spike test.

Appendix

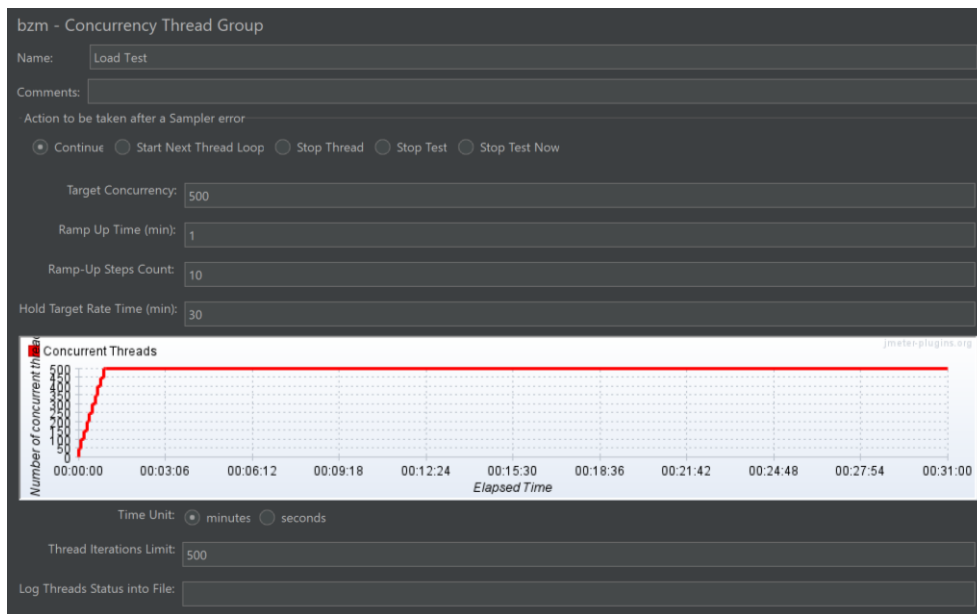
Load Test



ice-HRM-Data-01 -
Load Testjmx

Configurations

- **Assertion** – No Assertion Integrated.
- **Timer** - Synchronizing Timer
- **Thread Group** - ConcurrencyThread Group
- **Controller** - Random Controller
- **Reports** - Summary, Aggregate, jp@gc graphs View Results Table & View Results Tree
- **Config Elements** - User Defined Variables, Header Manager, Cookie & Cache Manager



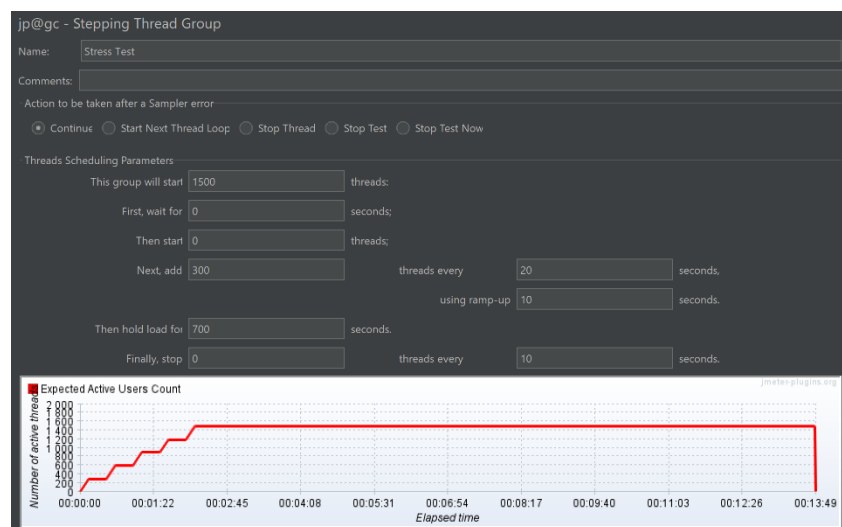
Stress Test



ice-HRM-Data-01 -
Stress Test.jmx

Configurations

- **Assertion** – No Assertion Added.
- **Timer** - Synchronizing Timer
- **Thread Group** - Stepping Thread Group
- **Controller** – Simple Controller
- **Reports** - Summary, Aggregate, jp@gc graphs View Results Table & View Results Tree
- **Config Elements** - User Defined Variables, Header Manager, Cookie & Cache Manager



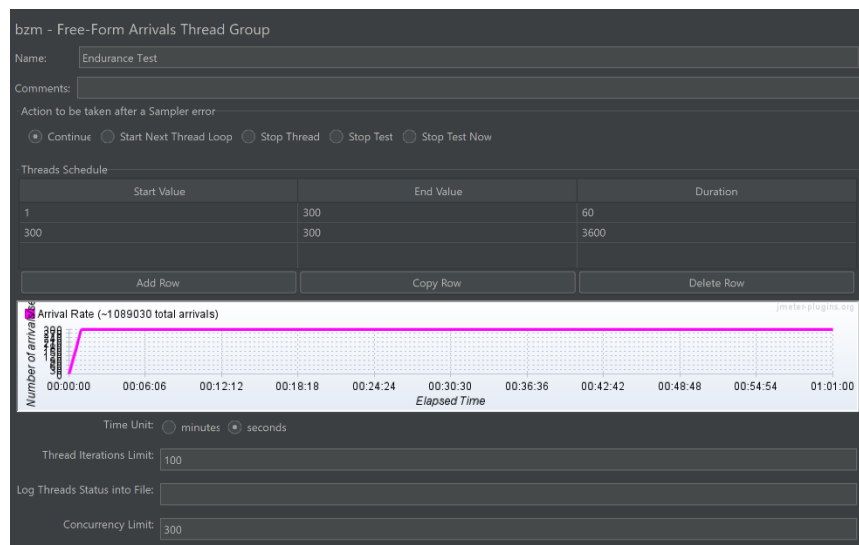
Endurance Test



ice-HRM-Data-01 -
Endurance Test.jmx

Configurations

- **Assertion** – No Assertion Integrated
- **Timer** - Synchronizing Timer
- **Thread Group** – Free-From Arrivals Thread Group
- **Controller** - Random Controller
- **Reports** - Summary, Aggregate, jp@gc graphs View Results Table & View Results Tree
- **Config Elements** - User Defined Variables, Header Manager, Cookie & Cache Manager



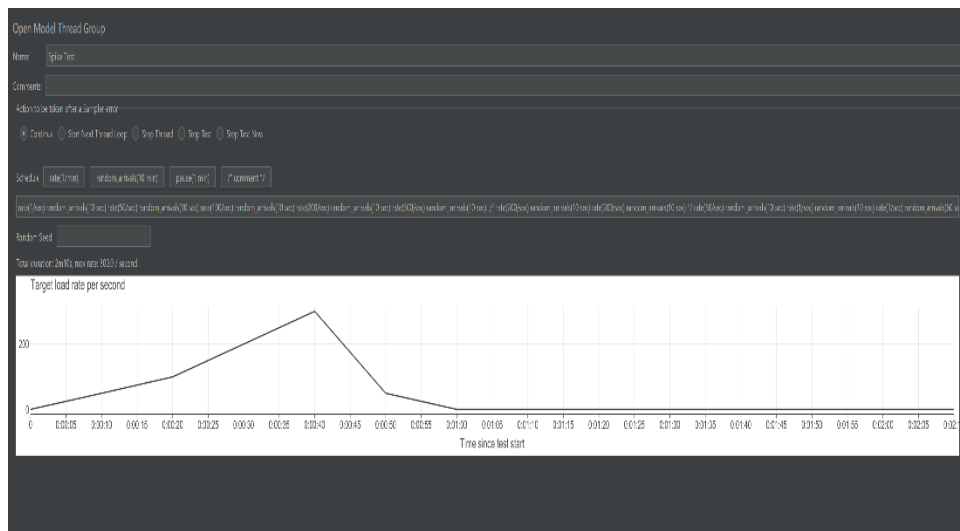
Spike Test



ice-HRM-Data-01 -
Spike Test.jmx

Configurations

- **Assertion** – No Assertion Integrated.
- **Timer** - Synchronizing Timer
- **Thread Group** – Open Model Thread Group
- **Controller** - Random Controller
- **Reports** - Summary, Aggregate, jp@gc graphs View Results Table & View Results Tree
- **Config Elements** - User Defined Variables, Header Manager, Cookie & Cache Manager



Volume Test



ice-HRM-Data-01 -
Spike Test.jmx

Configurations

- **Assertion** – No assertion added.
- **Timer** - Synchronizing Timer
- **Thread Group** - Ultimate Thread Group
- **Controller** - Random Controller
- **Reports** - Summary, Aggregate, jp@gc graphs View Results Table & View Results Tree
- **Config Elements** - User Defined Variables, Header Manager, Cookie & Cache Manager

